

EE133 Solid-State Electronics – Midterm

Winter 2026

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Given Constants:

- $n_i = 10^{10} \text{ cm}^{-3}$ at $T = 300 \text{ K}$
- $E_G = 1.12 \text{ eV}$ (Si), $\epsilon_s = 11.9 \epsilon_0$
- $kT = 0.026 \text{ eV}$ at $T = 300 \text{ K}$, $q = 1.602 \times 10^{-19} \text{ C}$
- $\epsilon_0 = 8.854 \times 10^{-14} \text{ F/cm}$, $m_0 = 9.11 \times 10^{-31} \text{ kg}$

Problem 1 – Miller Indices of the x-y Plane

(a) (10 pts) What are the Miller indices of the x-y plane?

The x-y plane is perpendicular to the z-axis. It has intercepts:

$$x \rightarrow \infty, \quad y \rightarrow \infty, \quad z \rightarrow 1$$

Taking the reciprocals:

$$\frac{1}{\infty} = 0, \quad \frac{1}{\infty} = 0, \quad \frac{1}{1} = 1$$

The Miller indices of the x-y plane are **(001)**.

(b) (10 pts) List all of the planes equivalent to the x-y plane.

The family of planes equivalent to (001) consists of all planes perpendicular to one of the three cubic axes. These are:

(001), (010), (100), (00 $\bar{1}$), (0 $\bar{1}$ 0), ($\bar{1}$ 00)

(c) (10 pts) Use the correct notation to indicate this full set of equivalent planes.

The full set of equivalent planes is denoted $\{001\}$ (curly braces notation).

Problem 2 – Miller Indices from the Diagram

(10 pts) What are the Miller indices of the plane at right?

From the diagram, the plane intercepts the axes at:

$$x = 3, \quad y = 4, \quad z = 5$$

Taking reciprocals:

$$\frac{1}{3}, \quad \frac{1}{4}, \quad \frac{1}{5}$$

Multiplying through by 60 (the LCM of 3, 4, 5) to get the smallest integers:

$$20, \quad 15, \quad 12$$

The Miller indices are **(20 15 12)**.

Problem 3 – Boron-Doped Silicon

Si is B-doped at $N_A = 10^{17} \text{ cm}^{-3}$, $T = 300 \text{ K}$.

(a) (10 pts) Determine the electron density.

Boron is a p-type (acceptor) dopant. Since $N_A = 10^{17} \gg n_i = 10^{10}$, we can assume complete ionization and charge neutrality gives $p \approx N_A = 10^{17} \text{ cm}^{-3}$.

Using the mass-action law:

$$n = \frac{n_i^2}{p} = \frac{(10^{10})^2}{10^{17}} = \frac{10^{20}}{10^{17}}$$

$$n = 10^3 \text{ cm}^{-3}$$

(b) (10 pts) Determine the hole density.

Since $N_A \gg n_i$:

$$p \approx N_A = 10^{17} \text{ cm}^{-3}$$

(c) (10 pts) Which best describes the material: (i) intrinsic or (ii) extrinsic? Why?

(ii) Extrinsic. The material is extrinsic because the dopant concentration $N_A = 10^{17} \text{ cm}^{-3}$ is many orders of magnitude larger than the intrinsic carrier concentration $n_i = 10^{10} \text{ cm}^{-3}$. The carrier concentrations are dominated by the Boron doping, not by thermal generation, so $p \gg n$.

(d) (20 pts) What is the value (in eV) of $(E_i - E_F)$?

For p-type material, E_F lies below E_i . We use:

$$p = n_i \exp\left(\frac{E_i - E_F}{kT}\right)$$

Solving for $E_i - E_F$:

$$\begin{aligned} E_i - E_F &= kT \ln\left(\frac{p}{n_i}\right) = 0.026 \text{ eV} \times \ln\left(\frac{10^{17}}{10^{10}}\right) = 0.026 \times \ln(10^7) \\ &= 0.026 \times 7 \times \ln(10) = 0.026 \times 7 \times 2.3026 = 0.026 \times 16.118 \end{aligned}$$

$$E_i - E_F = 0.419 \text{ eV}$$

Since this is positive, E_F is 0.419 eV below E_i , consistent with p-type material.

Problem 4 – Largest Intrinsic Carrier Concentration

(10 pts) At room temperature, which semiconductor has the largest intrinsic carrier concentration?

The intrinsic carrier concentration is given by:

$$n_i \propto \exp\left(-\frac{E_G}{2kT}\right)$$

A smaller bandgap E_G leads to a larger n_i because the exponential becomes less negative. Comparing the options:

- (a) Si: $E_G = 1.1$ eV
- (b) Ge: $E_G = 0.7$ eV
- (c) GaAs: $E_G = 1.4$ eV
- (d) InAs: $E_G = 0.35$ eV
- (e) InSb: $E_G = 0.17$ eV

(e) InSb ($E_G = 0.17$ eV) has the largest intrinsic carrier concentration because it has the smallest bandgap. Since $n_i \propto \exp(-E_G/2kT)$, the smallest E_G produces the largest exponential factor, thus the highest n_i .

Problem 5 – Highest Electron Mobility

(10 pts) At room temperature, which semiconductor has the highest electron mobility?

All three choices are Si at $T = 300$ K, but with different donor concentrations:

- (a) $N_D = 10^{14} \text{ cm}^{-3}$
- (b) $N_D = 10^{17} \text{ cm}^{-3}$
- (c) $N_D = 10^{18} \text{ cm}^{-3}$

Mobility is affected by both lattice scattering and ionized impurity scattering. As the doping concentration increases, there are more ionized impurities, which increases ionized impurity scattering and decreases mobility. At $N_D = 10^{14} \text{ cm}^{-3}$, the doping is low enough that lattice scattering dominates and mobility is near its maximum value.

(a) Si with $N_D = 10^{14} \text{ cm}^{-3}$ has the highest electron mobility. Lower doping means fewer ionized impurity scattering centers, so the mobility is highest. As N_D increases, ionized impurity scattering reduces μ_n .

Problem 6 – Highest Conductivity

(10 pts) Which of the above (from Q5) has the highest conductivity?

Conductivity for n-type material is:

$$\sigma = q n \mu_n$$

While option (a) has the highest mobility, option (c) has vastly more carriers. The carrier concentration increases by a factor of 10^4 from (a) to (c), while mobility decreases by roughly a factor of 2–3 at most. The product $n\mu_n$ is therefore largest for (c).

For (a): $\sigma \approx (1.602 \times 10^{-19})(10^{14})(\sim 1400) \approx 2.24 \times 10^{-2} \text{ S/cm}$

For (c): $\sigma \approx (1.602 \times 10^{-19})(10^{18})(\sim 250) \approx 40 \text{ S/cm}$

(c) Si with $N_D = 10^{18} \text{ cm}^{-3}$ has the highest conductivity. Although its mobility is lower, the much higher carrier concentration dominates since $\sigma = qn\mu_n$, and n increases far more than μ_n decreases.

Problem 7 – Band Diagram Analysis

From the band diagram: E_C , E_V , E_i , and E_F are all shown. The bands slope downward from left to right (i.e., E_C and E_V decrease with increasing x). Also, E_{Fn} and E_{Fp} are *not* equal – E_{Fn} is above E_{Fp} .

(a) (5 pts) It is in equilibrium. T / F

False. The quasi-Fermi levels for electrons (E_{Fn}) and holes (E_{Fp}) are split (they are not equal). In true thermal equilibrium, there is only one Fermi level, so the system is *not* in equilibrium.

(b) (5 pts) The electric field is pointing in which direction?

The electric field is related to the band bending by:

$$\mathcal{E} = \frac{1}{q} \frac{dE_C}{dx}$$

Since E_C decreases from left to right, $dE_C/dx < 0$, so $\mathcal{E} < 0$, meaning the field points in the negative x -direction (to the left).

(i) **Left.** The bands slope downward to the right ($dE_C/dx < 0$), so the electric field points to the left.

(c) (5 pts) There is an electron drift current flowing...

Electron drift current (conventional current) is:

$$J_{n,\text{drift}} = qn\mu_n\mathcal{E}$$

Since $q > 0$, $n > 0$, $\mu_n > 0$, and $\mathcal{E}$ points to the left, $J_{n,\text{drift}}$ flows to the left.

(i) **To the left.**

(d) (5 pts) There is an electron diffusion current flowing...

From the band diagram, E_C and E_{Fn} decrease at the same rate (they are parallel). This means $E_C - E_{Fn}$ is constant across position. Since $n \propto \exp(-(E_C - E_{Fn})/kT)$, the electron concentration is uniform, so $dn/dx = 0$.

$$J_{n,\text{diff}} = qD_n \frac{dn}{dx} = 0$$

(iii) **Neither, since it is zero.** E_C and E_{Fn} are parallel, so $E_C - E_{Fn}$ is constant, meaning the electron concentration is uniform and there is no diffusion current.

(e) (5 pts) There is a hole drift current flowing...

Hole drift current flows in the same direction as the electric field:

$$J_{p,\text{drift}} = qp\mu_p\mathcal{E}$$

Since $\mathcal{E}$ points to the left:

(i) To the left.

(f) (5 pts) There is a hole diffusion current flowing...

Hole concentration depends on $E_{Fp} - E_V$: $p \propto \exp(-(E_{Fp} - E_V)/kT)$. From the diagram, E_V decreases to the right while E_{Fp} is approximately flat, so $E_{Fp} - E_V$ *increases* to the right, meaning p *decreases* to the right. Holes diffuse from high concentration (left) to low concentration (right). The hole diffusion current is:

$$J_{p,\text{diff}} = -qD_p \frac{dp}{dx}$$

Since $dp/dx < 0$ (hole concentration decreases to the right), $J_{p,\text{diff}} = -qD_p(\text{negative}) > 0$, so the current flows to the right.

(ii) To the right.

(g) (5 pts) The total electron current is flowing...

The total electron current is proportional to the gradient of the electron quasi-Fermi level:

$$J_n = n\mu_n \frac{dE_{Fn}}{dx}$$

From the diagram, E_{Fn} slopes downward from left to right ($dE_{Fn}/dx < 0$). Therefore:

(i) To the left. Since E_{Fn} has a negative slope ($dE_{Fn}/dx < 0$), the total electron current flows in the negative x -direction (to the left).

(h) (5 pts) The total hole current is flowing...

Similarly:

$$J_p = -p\mu_p \frac{dE_{Fp}}{dx}$$

From the diagram, E_{Fp} appears approximately flat/constant ($dE_{Fp}/dx \approx 0$).

(iii) Neither, since it is zero. E_{Fp} is approximately constant (flat), so the total hole current is approximately zero.

Problem 8 – Minority Carrier Hole Distributions

The figure shows two minority carrier hole distributions (excess hole concentration δp_n vs. position x): a solid curve that decays exponentially and a dashed curve that decays linearly (steeper initial slope). Both start at the same δp_n value at $x = 0$ and decay to zero at $x = W$.

(a) (5 pts) Circle the one which results in the largest diffusion current at $x = 0$.

The **solid (exponential) curve** produces the largest diffusion current at $x = 0$ because it starts at the higher excess concentration δp_n and has the steeper slope (larger $|dp/dx|$) at $x = 0$.

(b) (5 pts) Explain using an equation.

The hole diffusion current is:

$$J_{p,\text{diff}} = -qD_p \left. \frac{dp}{dx} \right|_{x=0}$$

The magnitude of the diffusion current depends on $|dp/dx|$ at $x = 0$. The solid curve begins at the higher initial excess concentration ($\delta p_n > \Delta p_n$) and decays exponentially, giving it a steeper initial slope at $x = 0$:

$$\left| \frac{dp}{dx} \right|_{x=0, \text{ solid}} > \left| \frac{dp}{dx} \right|_{x=0, \text{ dashed}}$$

Since $|J_p| = qD_p |dp/dx|$, the solid curve's steeper slope at $x = 0$ gives a larger diffusion current.

(c) (5 pts) What direction is the hole diffusion current flowing?

Both curves show δp_n decreasing with increasing x (i.e., $dp/dx < 0$). Holes diffuse from high to low concentration, meaning holes move in the $+x$ direction. Since holes are positive carriers, the diffusion current flows in the same direction as the hole flow.

The hole diffusion current is flowing in the **positive x -direction** (to the right), from high concentration toward low concentration.

(d) (10 pts) Comparing the solid curve to the dashed curve, what physical mechanism is present or absent?

The four solutions of the minority carrier diffusion equation (MCDE) include terms with $\exp(-x/L_p)$, $\exp(+x/L_p)$, $\sinh$, and $\cosh$ forms, depending on boundary conditions.

The solid curve shows an exponential decay $\delta p_n \propto \exp(-x/L_p)$, which is the solution when recombination is present (the minority carrier lifetime τ_p is finite). The diffusion length $L_p = \sqrt{D_p \tau_p}$ governs the decay.

The dashed curve is linear, which is the solution when there is *no recombination* ($\tau_p \rightarrow \infty$, or $L_p \rightarrow \infty$). In this limit, the MCDE reduces to $d^2(\delta p)/dx^2 = 0$, giving a linear profile.

Recombination is the physical mechanism that differs between the two curves. The solid (exponential) curve has recombination present (τ_p is finite), causing the excess carriers to decay exponentially with diffusion length $L_p = \sqrt{D_p \tau_p}$. The dashed (linear) curve represents the case

where recombination is *absent* ($\tau_p \rightarrow \infty$), so the MCDE simplifies to $d^2(\delta p)/dx^2 = 0$, yielding a straight-line profile.

Problem 9 – Resistor Bar of p-type Si

Given: $L = 10 \mu\text{m}$, $W = 1 \mu\text{m}$, $t = 0.1 \mu\text{m}$, $N_A = 10^{18} \text{ cm}^{-3}$, $\mu_p = 642.2 \text{ cm}^2/\text{Vs}$, $m_p^* = 0.4 m_0$, $J = 1 \times 10^5 \text{ A/cm}^2$.

First, convert dimensions: $L = 10 \times 10^{-4} \text{ cm} = 10^{-3} \text{ cm}$, $W = 10^{-4} \text{ cm}$, $t = 10^{-5} \text{ cm}$.

The conductivity is:

$$\sigma = q N_A \mu_p = (1.602 \times 10^{-19})(10^{18})(642.2) = 0.1602 \times 642.2 = 102.88 \text{ S/cm}$$

The resistivity is:

$$\rho = \frac{1}{\sigma} = \frac{1}{102.88} = 9.72 \times 10^{-3} \Omega \cdot \text{cm}$$

(a) (10 pts) What is the sheet resistance?

$$R_{\text{sh}} = \frac{\rho}{t} = \frac{9.72 \times 10^{-3} \Omega \cdot \text{cm}}{10^{-5} \text{ cm}}$$

$$R_{\text{sh}} = 972 \Omega/\square$$

(b) (10 pts) What is the resistance?

Using $R = \rho L/A$ where $A = W \times t$:

$$A = W \times t = (10^{-4})(10^{-5}) = 10^{-9} \text{ cm}^2$$

$$R = \frac{\rho L}{A} = \frac{(9.72 \times 10^{-3})(10^{-3})}{10^{-9}} = \frac{9.72 \times 10^{-6}}{10^{-9}}$$

Or equivalently, $R = R_{\text{sh}} \times \frac{L}{W} = 972 \times \frac{10}{1}$:

$$R = 9720 \Omega \approx 9.72 \text{ k}\Omega$$

(c) (20 pts) What is the average drift velocity of a hole (cm/s)?

The current density is related to drift velocity by:

$$J = qp v_d$$

Since $p \approx N_A = 10^{18} \text{ cm}^{-3}$:

$$v_d = \frac{J}{qp} = \frac{1 \times 10^5}{(1.602 \times 10^{-19})(10^{18})} = \frac{10^5}{0.1602}$$

$$v_d = 6.24 \times 10^5 \text{ cm/s}$$

(d) (20 pts) For a hole, what is the average time between collisions (average scattering time)?

The mobility is related to the scattering time by:

$$\mu_p = \frac{q \tau_{\text{sc}}}{m_p^*} \implies \tau_{\text{sc}} = \frac{\mu_p m_p^*}{q}$$

First, $m_p^* = 0.4 m_0 = 0.4 \times 9.11 \times 10^{-31} = 3.644 \times 10^{-31}$ kg.

Converting μ_p to SI units: $642.2 \text{ cm}^2/\text{Vs} = 642.2 \times 10^{-4} \text{ m}^2/\text{Vs} = 6.422 \times 10^{-2} \text{ m}^2/\text{Vs}$.

$$\tau_{\text{sc}} = \frac{(6.422 \times 10^{-2})(3.644 \times 10^{-31})}{1.602 \times 10^{-19}} = \frac{2.340 \times 10^{-32}}{1.602 \times 10^{-19}}$$

$$\tau_{\text{sc}} = 1.46 \times 10^{-13} \text{ s}$$

Problem 10 – Illuminated Semiconductor Bar

Given: $N_D = 10^{16} \text{ cm}^{-3}$, $p(x=0) = 10^{12} \text{ cm}^{-3}$, $n_i = 10^{10} \text{ cm}^{-3}$, $\mu_n = 1154 \text{ cm}^2/\text{Vs}$, $\mu_p = 384.6 \text{ cm}^2/\text{Vs}$, $\tau_n = \tau_p = 0.1 \text{ }\mu\text{s} = 10^{-7} \text{ s}$, $T = 300 \text{ K}$.

First, find equilibrium values. Since $N_D = 10^{16} \gg n_i$:

$$n_0 = N_D = 10^{16} \text{ cm}^{-3}, \quad p_0 = \frac{n_i^2}{N_D} = \frac{10^{20}}{10^{16}} = 10^4 \text{ cm}^{-3}$$

The excess minority carrier concentration at $x=0$:

$$\delta p(0) = p(0) - p_0 = 10^{12} - 10^4 \approx 10^{12} \text{ cm}^{-3}$$

Diffusion coefficient for holes:

$$D_p = \frac{kT}{q} \mu_p = (0.026)(384.6) = 9.9996 \approx 10 \text{ cm}^2/\text{s}$$

Diffusion length for holes:

$$L_p = \sqrt{D_p \tau_p} = \sqrt{(10)(10^{-7})} = \sqrt{10^{-6}} = 10^{-3} \text{ cm} = 10 \text{ }\mu\text{m}$$

(a) (10 pts) What is the optical generation rate g_{op} in $(\text{cm}^{-3} \text{ s}^{-1})$?

At steady state, at $x=0$ the generation must sustain the excess carriers. The excess concentration satisfies:

$$\begin{aligned} \delta p(0) &= g_{\text{op}} \tau_p \\ g_{\text{op}} &= \frac{\delta p(0)}{\tau_p} = \frac{10^{12}}{10^{-7}} \end{aligned}$$

$$g_{\text{op}} = 10^{19} \text{ cm}^{-3} \text{ s}^{-1}$$

(b) (10 pts) Are the conditions of low-level injection satisfied? Why or why not?

Low-level injection requires $\delta p \ll n_0$ (excess minority carriers much less than the majority carrier concentration).

$$\begin{aligned} \delta p(0) &= 10^{12} \text{ cm}^{-3}, \quad n_0 = N_D = 10^{16} \text{ cm}^{-3} \\ \delta p(0) &= 10^{12} \ll 10^{16} = n_0 \quad \checkmark \end{aligned}$$

Yes, low-level injection conditions are satisfied because $\delta p(0) = 10^{12} \text{ cm}^{-3} \ll n_0 = 10^{16} \text{ cm}^{-3}$. The excess minority carrier concentration is four orders of magnitude smaller than the majority carrier concentration.

(c) (10 pts) Calculate the electron and hole densities at $x = 100 \text{ }\mu\text{m}$.

Since no light penetrates beyond $x = 0$, the excess holes decay exponentially for $x > 0$:

$$\delta p(x) = \delta p(0) \exp\left(-\frac{x}{L_p}\right)$$

At $x = 100 \mu\text{m}$:

$$\begin{aligned}\delta p(100 \mu\text{m}) &= 10^{12} \exp\left(-\frac{100}{10}\right) = 10^{12} e^{-10} \\ e^{-10} &= 4.54 \times 10^{-5} \\ \delta p &= 10^{12} \times 4.54 \times 10^{-5} = 4.54 \times 10^7 \text{ cm}^{-3}\end{aligned}$$

At $x = 100 \mu\text{m}$:

$$\begin{aligned}p &= p_0 + \delta p = 10^4 + 4.54 \times 10^7 \approx 4.54 \times 10^7 \text{ cm}^{-3} \\ n &= n_0 + \delta n = 10^{16} + 4.54 \times 10^7 \approx 10^{16} \text{ cm}^{-3}\end{aligned}$$

(The excess carriers are negligible compared to the majority electron concentration.)

(d) (20 pts) Calculate the hole current at $x = 0$.

Under low-level injection with no electric field in the quasi-neutral region, the minority carrier (hole) current density at $x = 0$ is purely diffusion:

$$J_p(x) = -qD_p \frac{d(\delta p)}{dx}$$

The excess carrier profile is $\delta p(x) = \delta p(0) e^{-x/L_p}$, so:

$$\left. \frac{d(\delta p)}{dx} \right|_{x=0} = -\frac{\delta p(0)}{L_p}$$

Therefore:

$$\begin{aligned}J_p(0) &= -qD_p \left(-\frac{\delta p(0)}{L_p} \right) = \frac{qD_p \delta p(0)}{L_p} \\ &= \frac{(1.602 \times 10^{-19})(10)(10^{12})}{10^{-3}} = \frac{1.602 \times 10^{-6}}{10^{-3}}\end{aligned}$$

$$J_p(0) \approx 1.6 \times 10^{-3} \text{ A/cm}^2 = 1.6 \text{ mA/cm}^2$$

The current flows in the positive x -direction (to the right), away from the illuminated surface, since holes diffuse from high concentration at $x = 0$ toward lower concentration.

(e) (10 pts) If we increase the recombination times, will the hole current increase or decrease? Why?

If τ_p increases, then $L_p = \sqrt{D_p \tau_p}$ increases. The current at $x = 0$ is:

$$J_p(0) = \frac{qD_p \delta p(0)}{L_p} = \frac{qD_p \delta p(0)}{\sqrt{D_p \tau_p}} = q \delta p(0) \sqrt{\frac{D_p}{\tau_p}}$$

As τ_p increases, $\sqrt{D_p/\tau_p}$ decreases, so $J_p(0)$ decreases.

The hole current will **decrease**. Increasing the recombination time increases the diffusion length L_p , which reduces the concentration gradient $|dp/dx|$ at $x = 0$. Since $J_p \propto 1/L_p \propto 1/\sqrt{\tau_p}$, a larger τ_p means a smaller hole current.

Problem 11 – Electrostatic Potential and Band Diagram

From the figure, the electrostatic potential $V(x)$ is:

- For $x < 0$: $V \approx 0$ V (constant)
- For $0 \leq x \leq 0.5 \mu\text{m}$: V decreases linearly from 0 V to -1 V
- For $x > 0.5 \mu\text{m}$: $V = -1$ V (constant)

(a) (10 pts) On plot (b), plot the conduction band edge.

The conduction band edge is related to the electrostatic potential by:

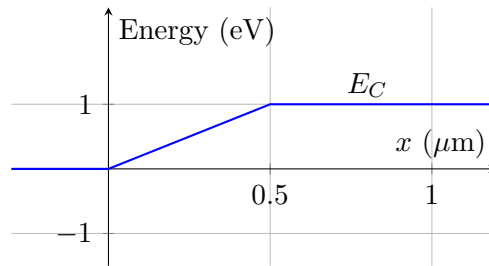
$$E_C(x) = E_{C,\text{ref}} - qV(x)$$

Since E_C is inversely related to V : where V goes down, E_C goes up, and vice versa.

From the diagram, plot (b) already has the start of E_C drawn at a constant level for $x < 0$. The conduction band should:

- For $x < 0$: E_C is constant at a reference level (since $V = 0$)
- For $0 \leq x \leq 0.5$: E_C increases linearly (since V decreases linearly, $-qV$ increases)
- For $x > 0.5$: E_C is constant at a higher value (since $V = -1$ V $\Rightarrow E_C = E_{C,\text{ref}} + 1$ eV)

The conduction band edge E_C mirrors the potential $V(x)$ with inverted sign ($E_C = \text{const} - qV$):



E_C is constant (low) for $x < 0$, increases linearly from $x = 0$ to $x = 0.5 \mu\text{m}$, then is constant (high) for $x > 0.5 \mu\text{m}$.

(b) (20 pts) Plot the electric field on plot (c).

The electric field is:

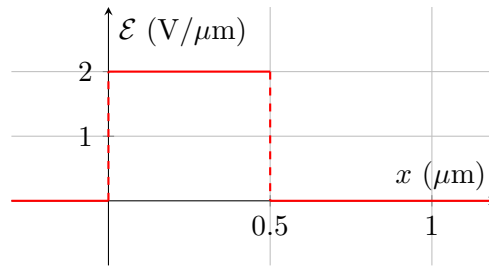
$$\mathcal{E} = -\frac{dV}{dx}$$

- For $x < 0$: V is constant, so $\mathcal{E} = 0$
- For $0 \leq x \leq 0.5 \mu\text{m}$: V decreases linearly from 0 to -1 V over $0.5 \mu\text{m}$:

$$\frac{dV}{dx} = \frac{-1 - 0}{0.5 \mu\text{m}} = -2 \text{ V}/\mu\text{m}$$

$$\mathcal{E} = -\frac{dV}{dx} = -(-2) = +2 \text{ V}/\mu\text{m}$$

- For $x > 0.5 \mu\text{m}$: V is constant, so $\mathcal{E} = 0$



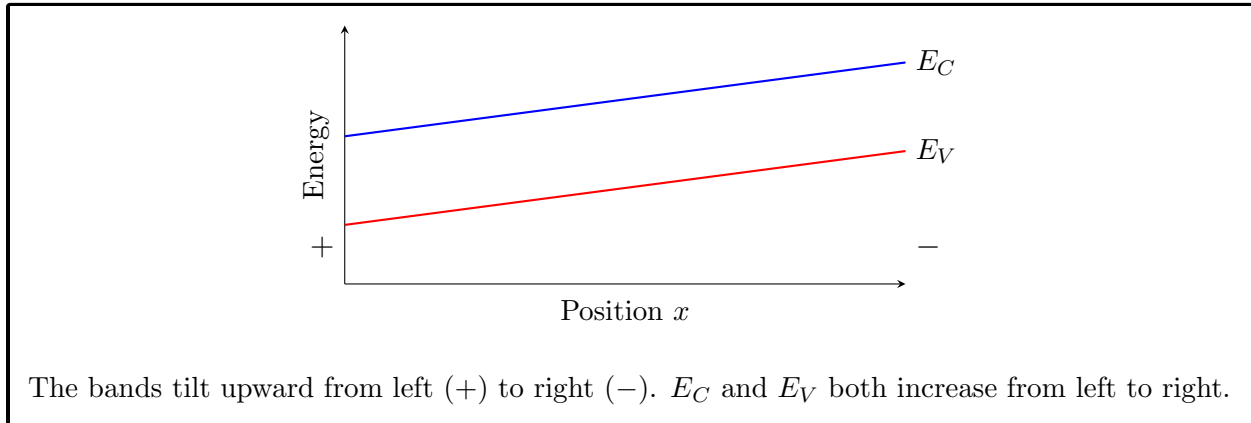
The electric field is $\mathcal{E} = 0$ for $x < 0$ and $x > 0.5 \mu\text{m}$, and $\mathcal{E} = +2 \text{ V}/\mu\text{m}$ (pointing in the $+x$ direction) for $0 < x < 0.5 \mu\text{m}$.

Problem 12 – Band Diagram with Applied Voltage

The band diagram of a uniformly doped n-type semiconductor (flat bands, electrons on E_C , holes on E_V) is drawn with $V = 0$. A positive voltage $V > 0$ is applied (+ on left, – on right).

(a) (10 pts) Sketch the band diagram with $V > 0$.

With $V > 0$ applied (positive terminal on the left), the potential on the left side is higher. Since $E_C = \text{const} - qV$, higher potential means lower band energy. The bands will tilt such that E_C and E_V are *lower on the left and higher on the right* (bands slope upward from left to right).



(b) (10 pts) Show directions of electron velocities v_n and hole velocities v_p .

The electric field points from high potential to low potential, i.e., from left (+) to right (–). So $\vec{\mathcal{E}}$ points to the right.

- Electrons (negative charge) drift *opposite* to $\vec{\mathcal{E}}$: v_n points to the **left**.
- Holes (positive charge) drift *with* $\vec{\mathcal{E}}$: v_p points to the **right**.

$v_n \leftarrow$ (electrons move to the left)
 $v_p \rightarrow$ (holes move to the right)

(c) (10 pts) Show which way the electric field is pointing.

$\vec{\mathcal{E}} \rightarrow$ The electric field points from the + terminal to the – terminal, i.e., to the **right**.

(d) (10 pts) Show which way the electrical current due to electrons, J_n , is flowing.

Electron drift current (conventional current) is opposite to electron velocity:

$$J_n = qn\mu_n\mathcal{E}$$

Since $\mathcal{E}$ is to the right and $q > 0$, $n > 0$, $\mu_n > 0$:

$J_n \rightarrow$ The electron current flows to the **right** (conventional current is opposite to electron motion).

(e) (10 pts) Show which way the electrical current due to holes, J_p , is flowing.

$$J_p = qp\mu_p\mathcal{E}$$

Hole current flows in the same direction as hole velocity (same as $\mathcal{E}$):

$J_p \rightarrow$ The hole current flows to the **right**. Both J_n and J_p flow in the same direction (to the right), so the total conventional current flows to the right.